CNN IMAGE CLASSIFICATION REPORT

Fashion-MNIST — 2D Convolutional Neural Network

Project overview
A convolutional neural network was implemented in PyTorch to classify 28 × 28 grayscale clothing images into ten Fashion-MNIST categories.

Item	Description
Dataset	Fashion-MNIST
Total images	70,000
Classes	10 clothing categories
Train / validation / test	54,000 / 6,000 / 10,000
Input shape	1 × 28 × 28
Device	CPU

1. Introduction

The objective was to build and evaluate a 2D CNN for image classification. Unlike tabular features, image pixels have spatial relationships. Convolutional layers exploit local patterns and share learned filters across image locations, making CNNs well suited to visual recognition tasks.

2. Dataset and preprocessing

Fashion-MNIST contains 70,000 grayscale images of size 28 × 28, split into ten balanced clothing categories. The official training set was divided into 54,000 training images and 6,000 validation images; the official test set of 10,000 images was kept separate. Images were normalized with mean 0.2860 and standard deviation 0.3530.

3. CNN architecture

Stage	Operation	Output shape
Input	Grayscale image	(1, 28, 28)
Block 1	Conv2D 1→32, 3×3, ReLU; MaxPool 2×2	(32, 14, 14)
Block 2	Conv2D 32→64, 3×3, ReLU; MaxPool 2×2	(64, 7, 7)
Block 3	Conv2D 64→128, 3×3, ReLU	(128, 7, 7)
Global pooling	Adaptive average pooling	(128,)
Classifier	Linear 128→64→10; ReLU; dropout 0.30	10 scores

There are 224 convolution filters in total (32 + 64 + 128). Global average pooling transforms the 128 feature maps into 128 values, which are passed to the dense classifier. The model contains 101,578 trainable parameters.

4. Training setup

Both experiments used Adam with learning rate 0.001 and batch size 128. The baseline allowed 12 epochs with early-stopping patience 4. The second experiment allowed 20 epochs with patience 5. The same architecture and data partitions were used for both runs.

5. Experiment comparison

Metric	12 epochs	20 epochs
Test accuracy	87.82%	89.71%
Macro precision	0.8786	0.8977
Macro recall	0.8782	0.8971
Macro F1-score	0.8778	0.8971
Macro ROC-AUC (OvR)	0.9898	0.9924
Best validation loss	0.3163	0.2694

The longer run improved test accuracy by 1.89 percentage points, corresponding to 189 additional correct classifications out of 10,000 test images. In the 20-epoch run, the best validation loss was 0.2694 at epoch 19, with validation accuracy of 90.45%. Validation loss increased slightly at epoch 20 to 0.2762.

6. Final per-class results — 20-epoch run

Class	Precision	Recall	F1	Support
T-shirt/top	0.86	0.81	0.84	1,000
Trouser	0.97	0.98	0.98	1,000
Pullover	0.85	0.86	0.86	1,000
Dress	0.87	0.92	0.89	1,000
Coat	0.85	0.80	0.82	1,000
Sandal	0.98	0.97	0.98	1,000
Shirt	0.70	0.73	0.71	1,000
Sneaker	0.94	0.97	0.96	1,000
Bag	0.98	0.97	0.98	1,000
Ankle boot	0.97	0.95	0.96	1,000

Shirt is the most difficult category (F1 = 0.71). The visually similar upper-body categories—Shirt, T-shirt/top, Coat and Pullover—account for many of the classification errors. Distinctive categories such as Trouser, Sandal, Bag, Sneaker and Ankle boot achieve stronger scores.

7. Training curves and evaluation plots

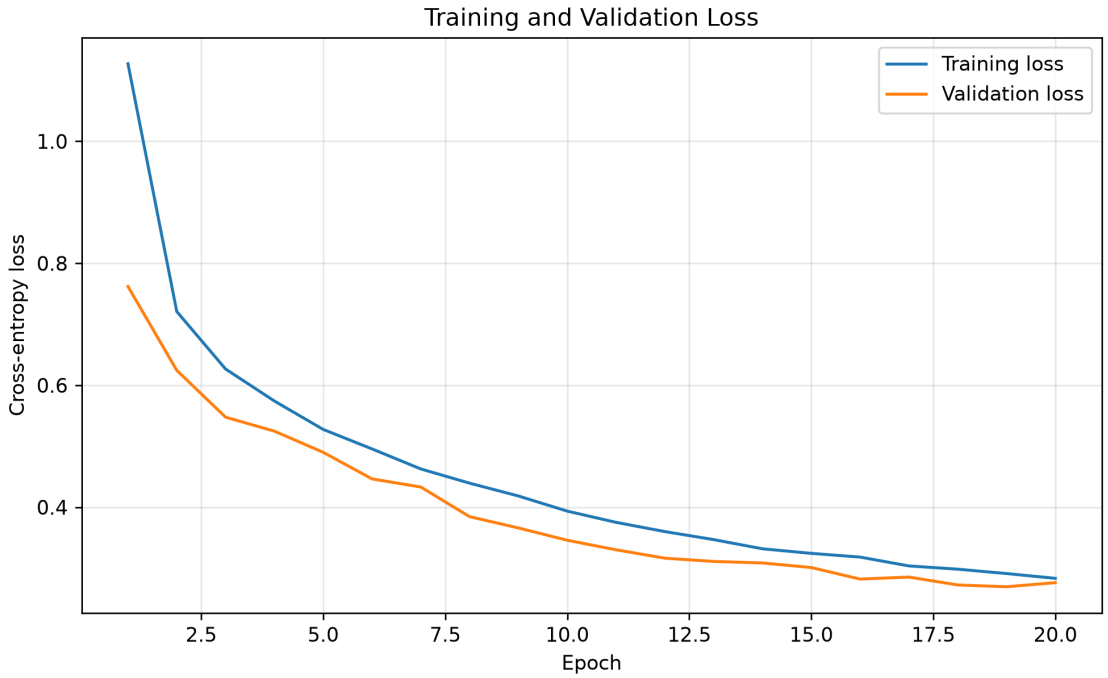


Figure 1. Training and validation loss for the 20-epoch run.

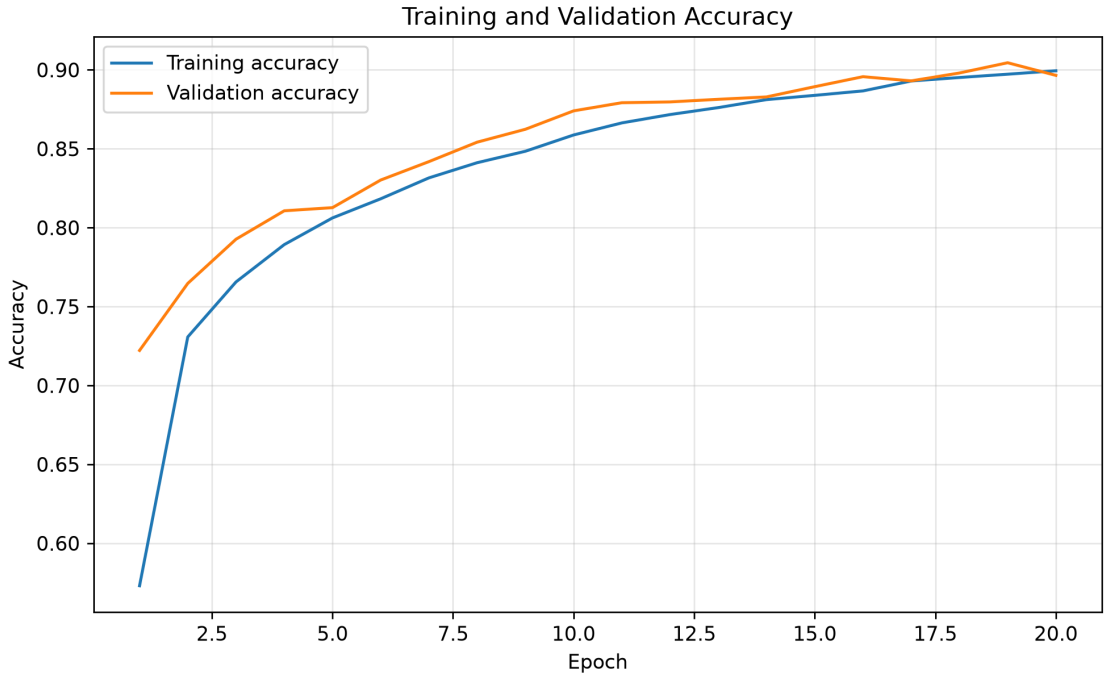


Figure 2. Training and validation accuracy.

Confusion matrix and feature visualization

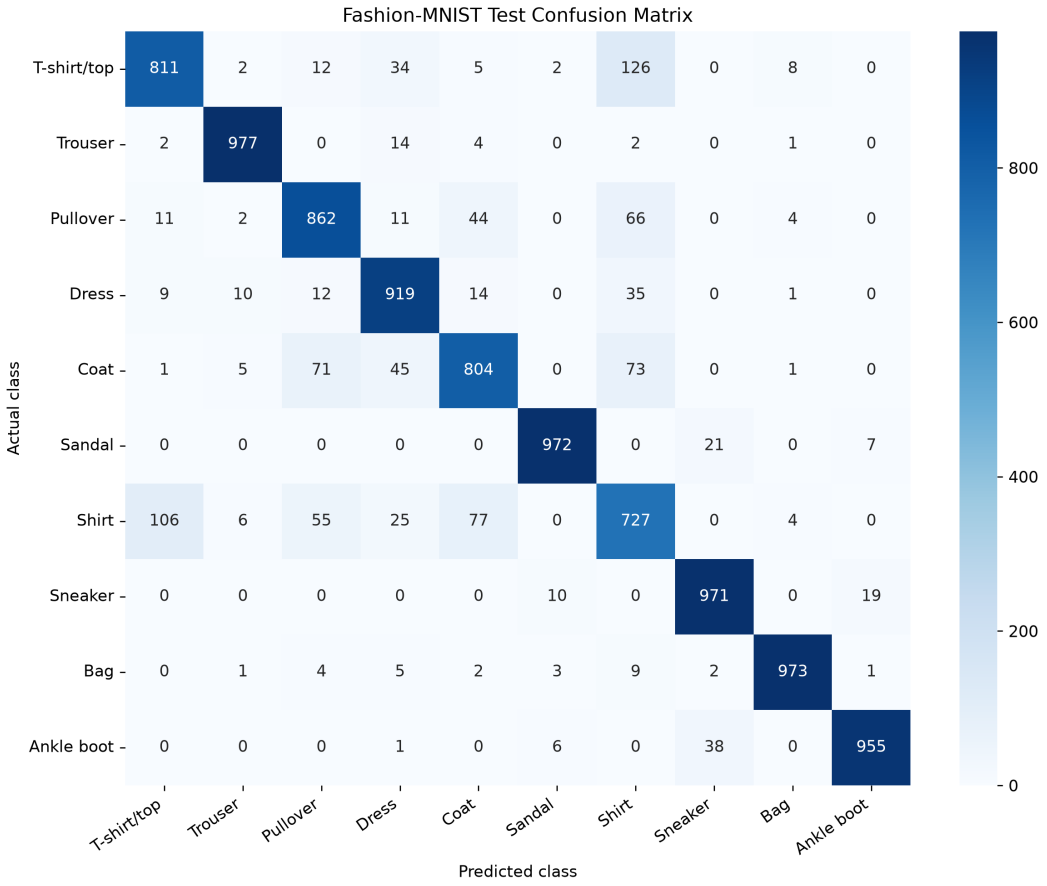


Figure 3. Confusion matrix evaluated on the 10,000-image test set.

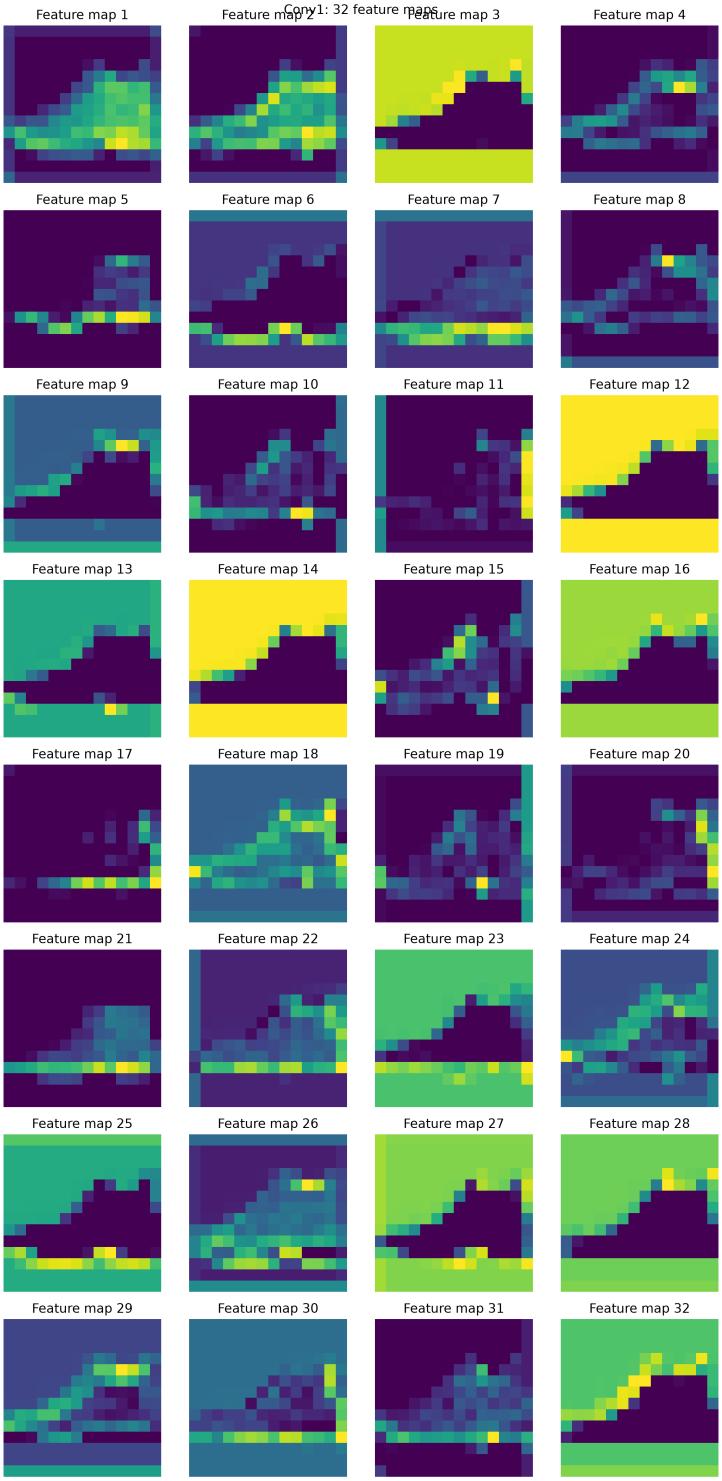


Figure 4. Feature maps produced by the first convolutional layer.

Learned filters and classifier representation

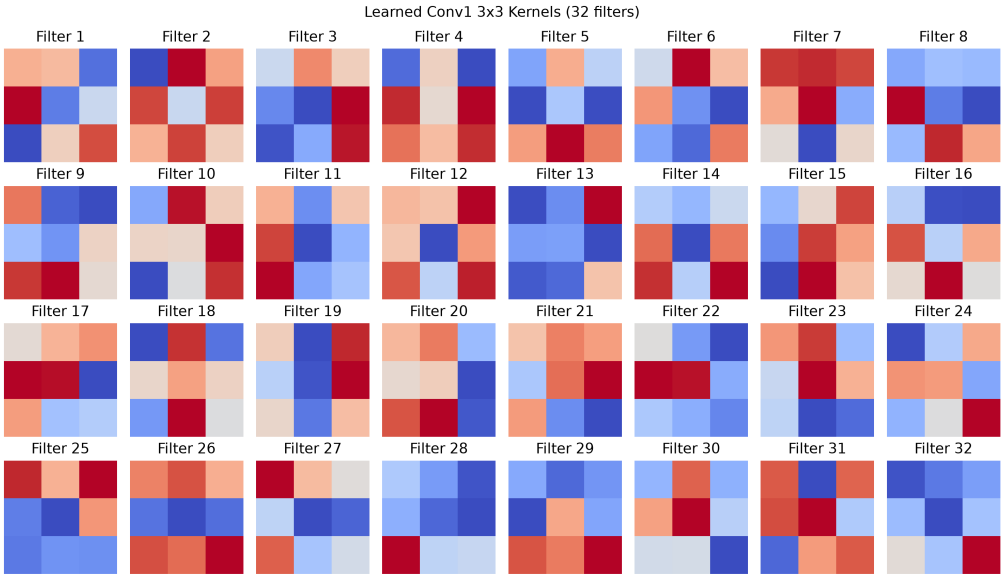


Figure 5. Learned kernels from the first convolutional layer.

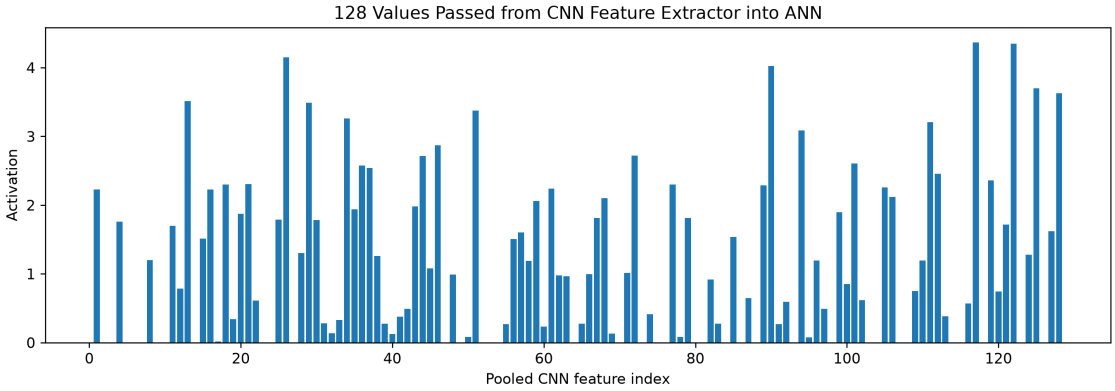


Figure 6. Example of the 128 values supplied to the ANN classifier after global average pooling.

8. Discussion

The CNN achieved 89.71% test accuracy and 0.8971 macro F1-score. The small gap between the 20-epoch validation accuracy at its best epoch (90.45%) and test accuracy (89.71%) is encouraging. The validation loss trend suggests the model was still learning through most of the run, although the slight increase at epoch 20 is a reminder to select checkpoints using validation performance.

ROC-AUC of 0.9924 is a one-vs-rest multiclass ranking metric; it is not the same as 99.24% classification accuracy.

9. Limitations and future work

- Implement an ANN baseline using flattened 784-pixel inputs and compare under the same split.
- Test mild image augmentation, such as small rotations and translations.
- Save and evaluate the checkpoint with the best validation loss rather than relying only on the final epoch.
- Repeat training with multiple random seeds to assess variability.
- Examine misclassified images and per-class confusion, especially for Shirt and other upper-body garments.

10. Conclusion

The project successfully implemented a three-stage Conv2D network for Fashion-MNIST. The final 20-epoch experiment achieved 89.71% test accuracy on 10,000 held-out images, improving on the 12-epoch baseline of 87.82%. The results provide a solid baseline for future experiments comparing ANN and CNN architectures and testing data augmentation.

Reproducibility note: Metrics and plots in this report are taken from the supplied 20-epoch output archive; baseline metrics are taken from the supplied baseline archive. Training was performed on CPU.